KIRLOSKAR CHILLERS PRIVATE LIMITED
A Kirloskar Group Company

Document Name	KWS, R-407C HEAT PUMP RATING CHART				
Document No.	ESP/14-15/06	Revision	R02	Date	28-Aug-17
Prepared by	AHW	Approved by	SSN	No. of sheets	5

KWS Ratings, R-407C Application

Unit Model	Compressor Model	Leaving Chilled Water Temp ^0C	Leaving Condenser Water Temperature ^0C								
			45			50			55		
			Cooling Capacity in TR	Power Input in KW	Heating Capacity in TR	Cooling Capacity in TR	Power Input in KW	Heating Capacity in TR	Cooling Capacity in TR	Power Input in KW	Heating Capacity in TR
KWS 055.17	CXH02 70 199Y	5.00	41.6	51.4	56.2	38.07	57.1	54.3	34.4	63.2	52.4
		6.00	42.9	51.9	57.7	39.25	57.7	55.7	35.5	63.8	53.6
		7.00	44.7	52.5	59.7	40.9	58.3	57.5	37.0	64.4	55.3
		8.00	46.1	52.9	61.1	42.2	58.8	58.9	38.1	65.0	56.6
		9.00	47.9	53.5	63.1	43.8	59.5	60.7	39.6	65.7	58.3
		10.00	49.7	54.0	65.0	45.4	60.1	62.5	41.1	66.4	60.0
KWS 070.17	CXH03 80 264Y	5.00	55.6	66.4	74.5	50.90	73.7	71.9	46.0	81.5	69.2
		6.00	57.3	67.0	76.4	52.47	74.5	73.7	47.5	82.3	70.9
		7.00	59.8	67.7	79.1	54.7	75.2	76.1	49.5	83.2	73.1
		8.00	61.6	68.3	81.0	56.4	75.9	78.0	51.0	83.9	74.8
		9.00	64.0	69.1	83.6	58.6	76.8	80.4	53.0	84.8	77.1
		10.00	66.4	69.7	86.2	60.8	77.5	82.8	54.9	85.7	79.3
KWS 095.17	CXH03 080 340Y	5.00	75.2	85.9	99.7	68.84	95.5	96.0	62.3	105.5	92.3
		6.00	77.6	86.8	102.2	70.97	96.4	98.4	64.2	106.6	94.5
		7.00	80.9	87.6	105.8	74.0	97.4	101.7	66.9	107.7	97.6
		8.00	83.3	88.4	108.5	76.2	98.3	104.2	68.9	108.6	99.8
		9.00	86.6	89.4	112.0	79.2	99.4	107.5	71.6	109.8	102.9
		10.00	89.8	90.3	115.5	82.2	100.3	110.7	74.3	110.9	105.8
KWS 110.17	CXH53 110 372Y	5.00	81.7	91.7	107.8	74.76	101.9	103.8	67.6	112.7	99.7
		6.00	84.2	92.6	110.6	77.07	103.0	106.4	69.7	113.8	102.1
		7.00	87.9	93.6	114.5	80.4	104.0	110.0	72.7	115.0	105.4
		8.00	90.5	94.4	117.3	82.8	105.0	112.7	74.9	116.0	107.9
		9.00	94.0	95.5	121.2	86.0	106.1	116.2	77.8	117.3	111.1
		10.00	97.5	96.4	124.9	89.2	107.1	119.7	80.7	118.4	114.4

Notes:
1) Cooling capacity and power input are based on $\Delta T = 5\ ^0C$ for entering/leaving chilled water temperature and $\Delta T = 5\ ^0C$ for entering/leaving condenser water
2) Total suction superheat is 5 ^0C and subcooling is 5 ^0C
3) Evaporator fouling factor = 0.0001 ft2hr^{0}F/BTU; Condenser fouling factor = 0.00025 ft2hr^{0}F/BTU

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KWS Ratings, R-407C Application

Unit Model	Compressor Model	Leaving Chilled Water Temp ^{0}C	Leaving Condenser Water Temperature ^{0}C								
			45			50			55		
			Cooling Capacity in TR	Power Input in KW	Heating Capacity in TR	Cooling Capacity in TR	Power Input in KW	Heating Capacity in TR	Cooling Capacity in TR	Power Input in KW	Heating Capacity in TR
KWS 135.17	CXH53 110 468Y	5.00	104.0	116.8	137.3	95.20	129.8	132.1	86.1	143.5	126.9
		6.00	107.2	118.0	140.8	98.14	131.1	135.4	88.7	144.9	130.0
		7.00	111.9	119.2	145.8	102.4	132.5	140.0	92.6	146.4	134.2
		8.00	115.2	120.3	149.4	105.4	133.7	143.5	95.3	147.7	137.4
		9.00	119.7	121.6	154.3	109.5	135.1	148.0	99.1	149.3	141.5
		10.00	124.2	122.8	159.1	113.6	136.4	152.4	102.8	150.8	145.6
KWS 180.17	CXH93 160 620Y	5.00	137.3	154.2	181.2	125.68	171.3	174.4	113.7	189.4	167.5
		6.00	141.6	155.7	185.9	129.56	173.1	178.8	117.2	191.3	171.6
		7.00	147.7	157.3	192.4	135.1	174.8	184.9	122.2	193.2	177.2
		8.00	152.1	158.7	197.3	139.2	176.4	189.4	125.9	195.0	181.3
		9.00	158.0	160.5	203.7	144.6	178.3	195.3	130.8	197.1	186.8
		10.00	163.9	162.0	210.0	150.0	180.1	201.2	135.7	199.0	192.3
KWS 205.17	CXH93 160 702Y	5.00	156.6	174.1	206.1	143.32	193.5	198.3	129.6	213.8	190.4
		6.00	161.5	175.8	211.5	147.75	195.4	203.3	133.6	216.0	195.0
		7.00	168.4	177.6	218.9	154.1	197.4	210.3	139.4	218.2	201.4
		8.00	173.5	179.2	224.4	158.7	199.2	215.4	143.5	220.2	206.2
		9.00	180.2	181.2	231.7	164.9	201.4	222.2	149.1	222.6	212.4
		10.00	186.9	183.0	239.0	171.1	203.3	228.9	154.7	224.7	218.6
KWS 230.17	CXH93 210 810Y	5.00	180.6	197.2	236.7	165.27	219.1	227.6	149.5	242.2	218.3
		6.00	186.2	199.1	242.8	170.37	221.3	233.3	154.1	244.6	223.6
		7.00	194.2	201.2	251.4	177.7	223.6	241.3	160.7	247.1	231.0
		8.00	200.0	203.0	257.8	183.0	225.6	247.2	165.5	249.4	236.4
		9.00	207.8	205.2	266.2	190.1	228.1	255.0	171.9	252.1	243.6
		10.00	215.6	207.2	274.5	197.3	230.3	262.8	178.4	254.6	250.8

Notes:
1) Cooling capacity and power input are based on ΔT = 5 ^{0}C for entering/leaving chilled water temperature and ΔT = 5 ^{0}C for entering/leaving condenser water
2) Total suction superheat is 5 ^{0}C and subcooling is 5 ^{0}C
3) Evaporator fouling factor = 0.0001 ft2hr^{0}F/BTU; Condenser fouling factor = 0.00025 ft2hr^{0}F/BTU

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KWS Ratings, R-407C Application

Unit Model	Compressor Model	Leaving Chilled Water Temp °C	Leaving Condenser Water Temperature °C								
			45			50			55		
			Cooling Capacity in TR	Power Input in KW	Heating Capacity in TR	Cooling Capacity in TR	Power Input in KW	Heating Capacity in TR	Cooling Capacity in TR	Power Input in KW	Heating Capacity in TR
KWS 110.27	CXH02 70 199Y X 2	5.00	83.3	102.9	112.5	76.20	114.4	108.7	68.9	126.4	104.9
		6.00	85.8	103.9	115.4	78.55	115.5	111.4	71.0	127.7	107.3
		7.00	89.5	105.0	119.4	81.9	116.7	115.1	74.1	129.0	110.8
		8.00	92.2	106.0	122.4	84.4	117.8	117.9	76.3	130.1	113.3
		9.00	95.8	107.1	126.3	87.7	119.0	121.5	79.3	131.6	116.7
		10.00	99.4	108.2	130.1	90.9	120.2	125.1	82.2	132.9	120.0
KWS 130.27	CXH02 70 199Y + CXH03 80 264Y	5.00	97.3	117.8	130.8	89.04	131.0	126.3	80.5	144.7	121.7
		6.00	100.3	119.0	134.2	91.79	132.3	129.4	83.0	146.2	124.6
		7.00	104.6	120.2	138.8	95.7	133.6	133.8	86.6	147.7	128.6
		8.00	107.8	121.3	142.3	98.6	134.8	137.0	89.2	149.0	131.6
		9.00	112.0	122.6	146.8	102.4	136.3	141.2	92.6	150.7	135.5
		10.00	116.1	123.8	151.4	106.3	137.6	145.4	96.1	152.1	139.4
KWS 150.27	CXH03 80 264Y + CXH03 80 298Y	5.00	118.7	140.0	158.6	108.66	155.6	152.9	98.3	172.0	147.2
		6.00	122.4	141.4	162.6	112.01	157.1	156.7	101.3	173.7	150.7
		7.00	127.7	142.9	168.3	116.8	158.8	162.0	105.7	175.5	155.6
		8.00	131.5	144.1	172.5	120.3	160.2	165.9	108.8	177.1	159.2
		9.00	136.6	145.7	178.1	125.0	161.9	171.1	113.0	179.0	164.0
		10.00	141.7	147.1	183.6	129.7	163.5	176.2	117.3	180.7	168.7

Notes:
1) Cooling capacity and power input are based on $\Delta T = 5\ ^0C$ for entering/leaving chilled water temperature and $\Delta T = 5\ ^0C$ for entering/leaving condenser water
2) Total suction superheat is 5 ^0C and subcooling is 5 ^0C
3) Evaporator fouling factor = 0.0001 ft2hr^{0}F/BTU; Condenser fouling factor = 0.00025 ft2hr^{0}F/BTU

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Unit Model	Compressor Model	Leaving Chilled Water Temp ^{0}C	Leaving Condenser Water Temperature ^{0}C								
			45			50			55		
			Cooling Capacity in TR	Power Input in KW	Heating Capacity in TR	Cooling Capacity in TR	Power Input in KW	Heating Capacity in TR	Cooling Capacity in TR	Power Input in KW	Heating Capacity in TR
KWS 180.27	CXH03 80 298Y + CXH03 080 340Y	5.00	138.3	159.5	183.6	126.52	177.3	176.9	114.4	196.0	170.1
		6.00	142.5	161.1	188.3	130.42	179.1	181.3	117.9	197.9	174.2
		7.00	148.7	162.8	195.0	136.0	180.9	187.5	123.0	200.0	179.9
		8.00	153.1	164.2	199.8	140.1	182.5	192.0	126.7	201.8	184.1
		9.00	159.1	166.0	206.3	145.6	184.5	198.0	131.6	204.0	189.6
		10.00	165.0	167.7	212.7	151.0	186.3	204.0	136.6	206.0	195.1
KWS 215.27	CXH53 110 372Y X 2	5.00	163.5	183.5	215.6	149.58	203.9	207.6	135.3	225.4	199.4
		6.00	168.5	185.3	221.2	154.19	205.9	212.8	139.4	227.6	204.2
		7.00	175.8	187.2	229.0	160.8	208.1	220.0	145.4	230.0	210.9
		8.00	181.0	188.9	234.8	165.7	209.9	225.4	149.8	232.0	215.8
		9.00	188.1	191.0	242.4	172.1	212.2	232.5	155.6	234.6	222.3
		10.00	195.1	192.8	249.9	178.5	214.3	239.5	161.4	236.9	228.8
KWS 270.27	CXH53 110 468Y X 2	5.00	208.1	233.6	274.5	190.40	259.6	264.2	172.2	287.0	253.8
		6.00	214.5	235.9	281.6	196.28	262.2	270.9	177.5	289.8	259.9
		7.00	223.7	238.4	291.5	204.7	264.9	280.1	185.1	292.8	268.4
		8.00	230.4	240.5	298.9	210.9	267.3	286.9	190.7	295.4	274.7
		9.00	239.4	243.1	308.6	219.1	270.2	295.9	198.1	298.7	283.0
		10.00	248.3	245.5	318.2	227.3	272.9	304.9	205.5	301.6	291.3

Notes:
1) Cooling capacity and power input are based on ΔT = 5 ^{0}C for entering/leaving chilled water temperature and ΔT = 5 ^{0}C for entering/leaving condenser water
2) Total suction superheat is 5 ^{0}C and subcooling is 5 ^{0}C
3) Evaporator fouling factor = 0.0001 ft2hr^{0}F/BTU; Condenser fouling factor = 0.00025 ft2hr^{0}F/BTU

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Unit Model	Compressor Model	Leaving Chilled Water Temp ^{0}C	Leaving Condenser Water Temperature ^{0}C								
			45			50			55		
			Cooling Capacity in TR	Power Input in KW	Heating Capacity in TR	Cooling Capacity in TR	Power Input in KW	Heating Capacity in TR	Cooling Capacity in TR	Power Input in KW	Heating Capacity in TR
KWS 315.27	CXH53 110 468Y + CXH93 160 620Y	5.00	241.3	271.0	318.4	220.85	301.2	306.5	199.7	332.9	294.4
		6.00	248.8	273.7	326.6	227.66	304.2	314.2	205.9	336.2	301.5
		7.00	259.5	276.5	338.2	237.5	307.3	324.9	214.7	339.7	311.3
		8.00	267.3	279.0	346.6	244.6	310.1	332.8	221.2	342.7	318.7
		9.00	277.7	282.0	357.9	254.1	313.4	343.2	229.8	346.4	328.3
		10.00	288.1	284.8	369.1	263.6	316.5	353.6	238.4	349.8	337.9
KWS 360.27	CXH93 160 620Y X 2	5.00	274.6	308.3	362.3	251.29	342.7	348.8	227.2	378.8	335.0
		6.00	283.1	311.4	371.7	259.05	346.1	357.5	234.3	382.6	343.1
		7.00	295.3	314.6	384.8	270.2	349.7	369.7	244.3	386.5	354.3
		8.00	304.1	317.5	394.4	278.3	352.8	378.7	251.7	390.0	362.6
		9.00	316.0	320.9	407.2	289.1	356.7	390.6	261.5	394.2	373.6
		10.00	327.8	324.1	419.9	299.9	360.2	402.4	271.2	398.1	384.4
KWS 385.27	CXH93 160 620Y + CXH93 160 702Y	5.00	293.9	328.3	387.3	268.98	364.8	372.7	243.2	403.2	357.9
		6.00	303.0	331.5	397.3	277.28	368.5	382.1	250.7	407.2	366.6
		7.00	316.1	335.0	411.3	289.2	372.3	395.1	261.5	411.4	378.6
		8.00	325.5	338.0	421.7	297.9	375.6	404.7	269.4	415.1	387.5
		9.00	338.2	341.7	435.4	309.5	379.7	417.5	279.9	419.7	399.2
		10.00	350.8	345.0	449.0	321.0	383.4	430.1	290.3	423.8	410.8

Notes:
1) Cooling capacity and power input are based on ΔT = 5 ^{0}C for entering/leaving chilled water temperature and ΔT = 5 ^{0}C for entering/leaving condenser water
2) Total suction superheat is 5 ^{0}C and subcooling is 5 ^{0}C
3) Evaporator fouling factor = 0.0001 ft2hr^{0}F/BTU; Condenser fouling factor = 0.00025 ft2hr^{0}F/BTU